PetroChina · KUNLUN Lubricant | TRANSFORMER OIL

KUNLUN Petro 45X Plus

Naphthenic insulating oil — Inhibited (contains metal passivant)

PRODUCT DATA SHEET

Product description

Petro 45X Plus Transformer Oil is obtained by adding a high-quality antioxidant additive to deeply-refined, low-viscosity, low-pour-point, low-sulphur-content naphthenic base oil produced by a special process. It offers rapid heat transfer, good oxidation stability, excellent electrical properties and good gassing resistance. The product is mainly exported.

Quality grade and standards

Quality grade: Highest grade.

Quality indexes according to IEC 60296:2012, Class I-30 °C (special purpose). Also meets ASTM D3487-16 Type II.

Application range

Suitable for distribution transformers, generation transformers, power transformers and reactors with a Lowest Cold Start Energizing Temperature (LCSET) of -30 °C, and oil-filled electrical equipment with similar requirements. The product contains metal passivants.

Properties & characteristics

– Relatively low viscosity and viscosity index for effective cooling and heat transfer
– Excellent heat and oxidation stability to prolong equipment life and reduce maintenance cost
– Excellent electrical insulation: high breakdown voltage and low dielectric loss factor
– Good gassing resistance to prevent air-gap discharge under high-voltage conditions
– Good low-temperature properties; free of pour-point depressant; pour point below -40 °C
– Free of DBDS; additives listed per IEC 60296:2012 and ASTM D3487-16; free of PCB

Typical properties

Quality indexes and typical data. Quality indexes according to IEC 60296:2012, Class I-30 °C (special purpose). Also meets ASTM D3487-16 Type II.

Property	Unit	Method	Min.	Max.	Typical data
Function characteristics (IEC 60296:2012)					
Pour point	°C	ISO 3016		-40	-58
Kinematic viscosity (40 °C)	mm²/s	ISO 3104		12	9.2
Kinematic viscosity (-30 °C)	mm²/s	ISO 3104		1800	1230
Water content	mg/kg	IEC 60814		30	20
Breakdown voltage – untreated	kV	IEC 60156	30		40–60
Breakdown voltage – treated	kV	IEC 60156	70		>70
Density (20 °C)	kg/m³	ISO 12185		895	876
Dielectric loss factor (90 °C)		IEC 60247		0.005	<0.001
Refining / stability characteristics					
Appearance		Visual		Clear & transparent	Clear & transparent
Acid number	mgKO	IEC 62021-1		0.01	<0.01

Property	Unit	Method	Min.	Max.	Typical data
	H/g				
Interfacial tension	mN/m	ASTM D971	40		46
Total sulphur content	%	ASTM D5453		0.15	<0.01
Corrosive sulphur		ASTM D1275B		Not corrosive	Not corrosive
Latent corrosive sulphur		IEC 62535		Not corrosive	Not corrosive
Corrosive sulphur		DIN 51353		Not corrosive	Not corrosive
DBDS content	mg/kg	IEC 62697-1		Not detectabl e	Not detectable
Antioxidant content	%	IEC 60666	0.08	0.40	0.29
Operation characteristics – Oxidation stability (120 °C, 500 h)					
Total acid number	mgKOH/g	IEC 61125C		0.3	0.10
Oil sludge	%	IEC 61125C		0.05	0.002
Dielectric loss factor (90 °C)		IEC 60247		0.050	0.035
Health, safety and environment					
Flash point (closed)	°C	ISO 2719	135		150
PCA content	%	IP 346		3	<3
PCB content	mg/kg	IEC 61619		Not detectabl e	Not detectable
ASTM D3487-16 (Type II) certificate – typical data					
Aniline point	°C	ASTM D611	63		82
Colour		ASTM D1500		0.5	<0.5
Flash point	°C	ASTM D92	145		160
Kinematic viscosity (0 °C)	cSt	ASTM D445		76.0	65
BDV – VDE electrode (1 mm)	kV	ASTM D1816	20		25
BDV – VDE electrode (2 mm)	kV	ASTM D1816	35		40
Impulse breakdown voltage	kV	ASTM D3300	145		390
Rotating bomb oxidation	min	ASTM D2112	195		378

Note: Typical data correspond to current production and are provided for reference; they are not a specification limit. The product is produced under a strict quality control procedure with constant product resources. Transformer oil is vulnerable to moisture and contamination – keep dry and clean, store in dedicated containers indoors or in a controlled climate, and keep free of acetylene gas.